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PAPER TITLE

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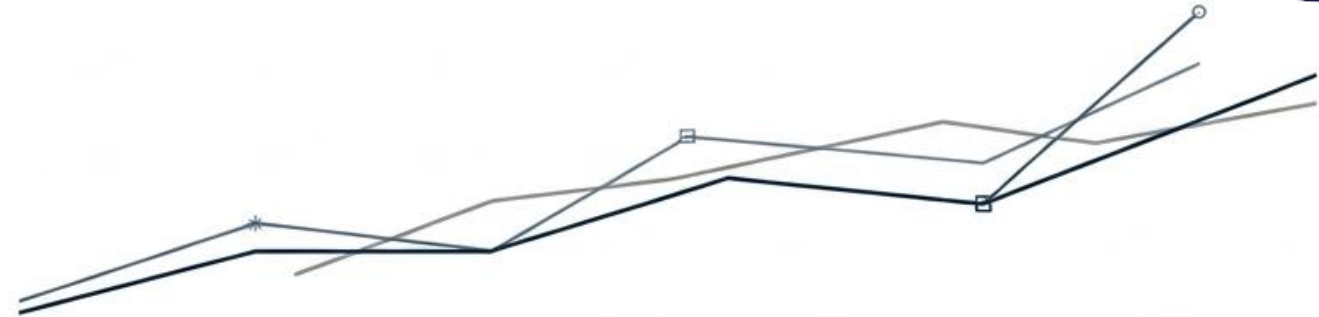
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OUTLINE



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Dilemma,
Economic Costs
& Evolution.

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Metrics &
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Standardisation.



MOTIVATION



MOTIVATION

The Cost of Unplanned Downtime

Aviation



20%

Maintenance budget lost to unscheduled, reactive repairs.



+5% fuel consumption penalty from sub-optimal component health.



Maritime



>50%

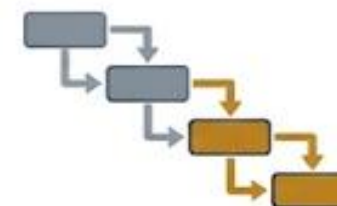
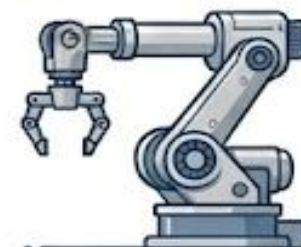
Proportion of maritime incidents in 2023 explicitly attributed to equipment malfunction.



11,500+ technical failure incidents reported between 2014-2023.



Automotive



Cascading Stoppages



Unplanned stoppages cause supply chain halts; currently the costliest operational risk in automated manufacturing.



MOTIVATION



MOTIVATION

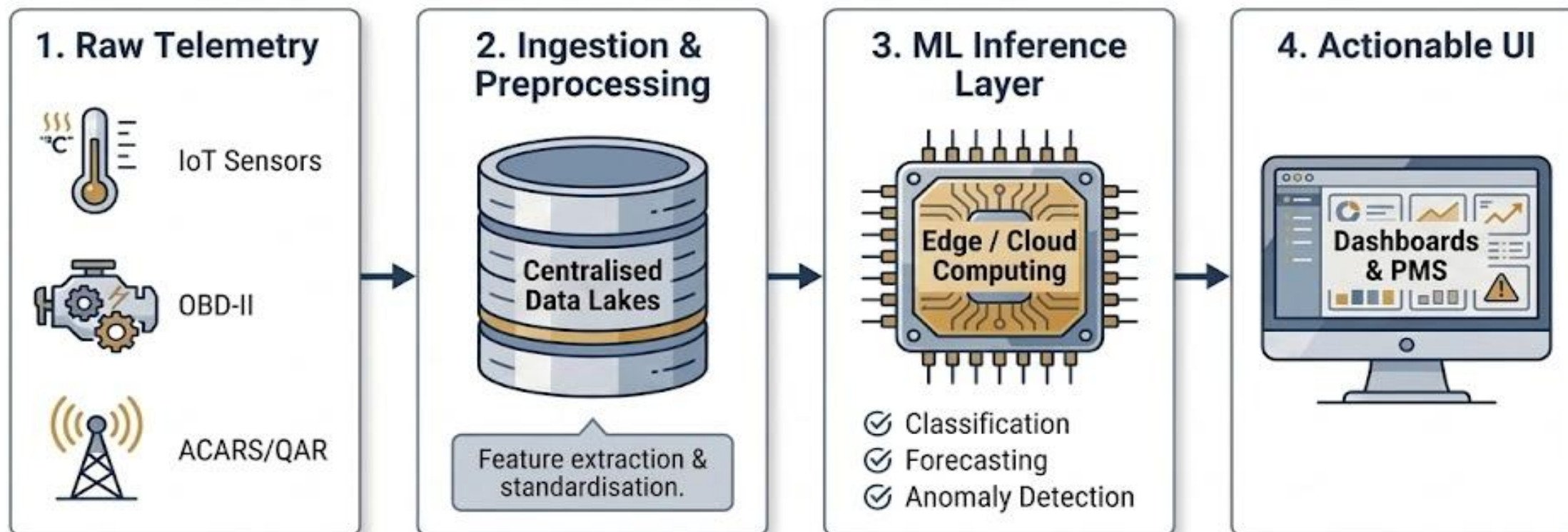
The Evolution of Maintenance Maturity

Corrective (Reactive)	Preventive (Time-Based)	Predictive (AI-Driven)
Failure	Scheduled Intervention	Intervention Point Critical failure Threshold
Concept: Fix when broken.	Concept: Replace on schedule.	Concept: Right on time.
Data: Zero proactive insight.	Data: Physics-based assumptions; static intervals.	Data: Real-time sensor telemetry & ML mapping functions.
Outcome: Maximum emergency cost; highest safety risk.	Outcome: Severe over-maintenance or missed pre-failure windows.	Outcome: Interventions triggered precisely by degradation data.



METHODS AND MATERIALS

The Universal PdM Architecture

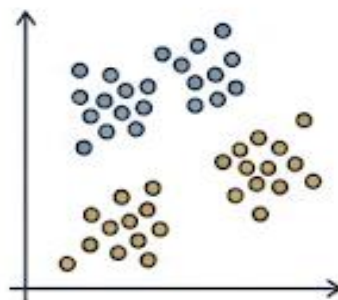


A universal architecture maps raw operational physics into actionable maintenance insights.



METHODS AND MATERIALS

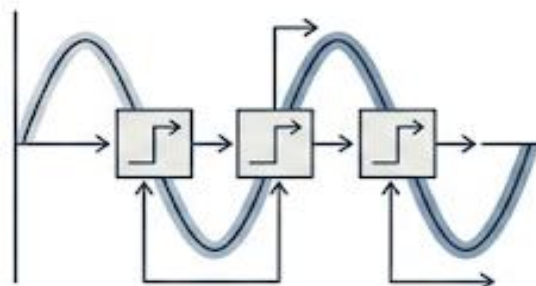
The AI Algorithmic Toolkit



Supervised Classification

kNN, SVM, Random Forest (RF)

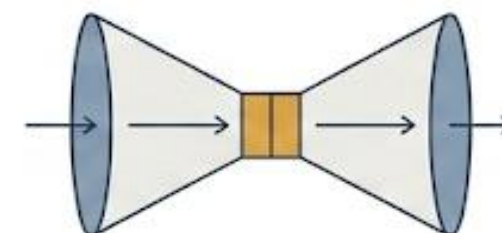
Mapping discrete fault states to identify exactly which system failure mode is currently occurring based on labeled historical data.



Time-Series Forecasting

Long Short-Term Memory (LSTM)

Predicting Remaining Useful Life (RUL) along a temporal trajectory by overcoming vanishing gradients via specialized memory gates.



Unsupervised Detection

Autoencoders

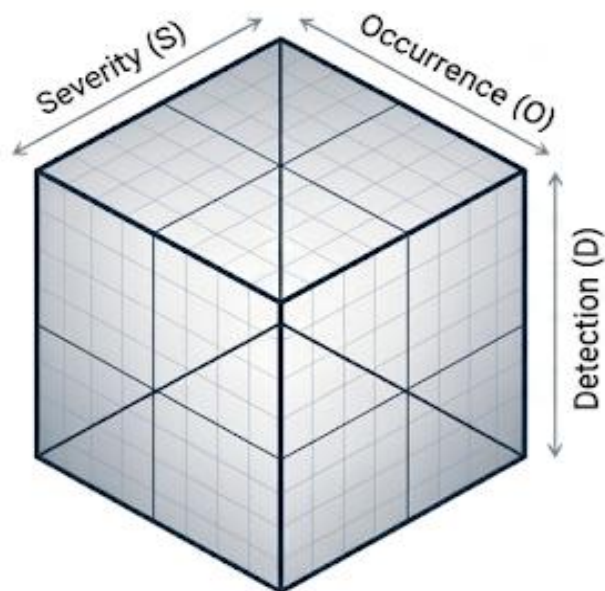
Flagging novel anomalies and deviations in complex manufacturing environments where labeled failure data is scarce or unavailable.



METHODS AND MATERIALS

Mathematical Foundations: Prioritisation & Classification

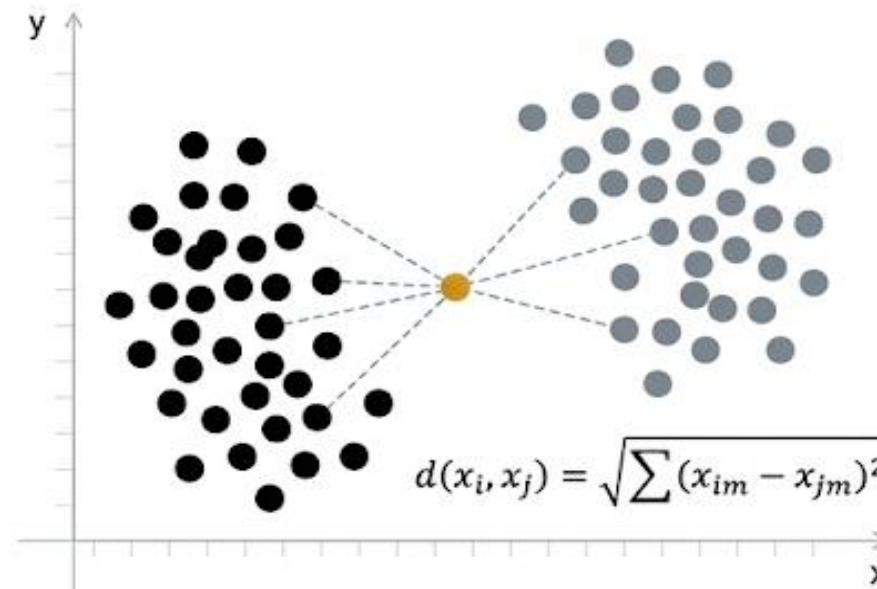
FMECA RISK PRIORITISATION



$$RPN = S \times O \times D$$

Higher Risk Priority Number (RPN) directs the ML classifier's training priorities.

KNN CLASSIFICATION



$$\hat{y} = \operatorname{argmax}_c |\{x_j \in N_k(x) : y_j = c\}|$$

Standardised features (zero mean, unit variance) evaluated at optimal $k=5$ via Euclidean distance.



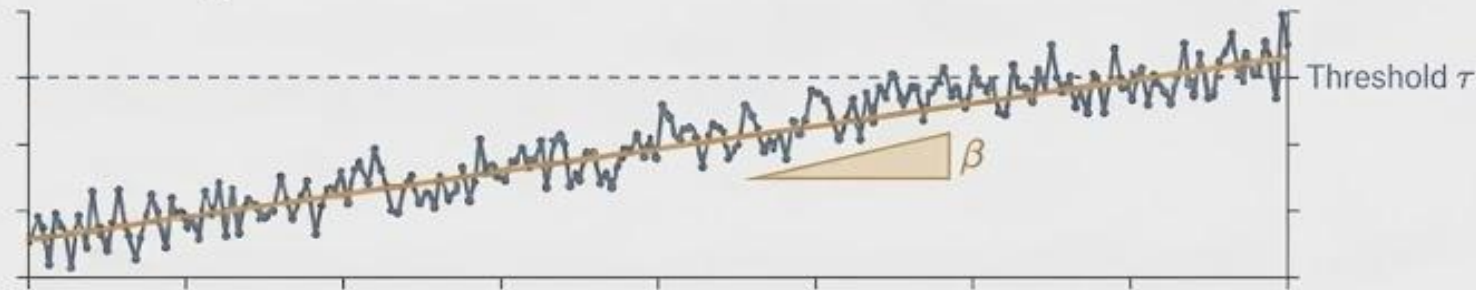
METHODS AND MATERIALS



METHODS AND MATERIALS

Mathematical Foundations: Degradation & Normalisation

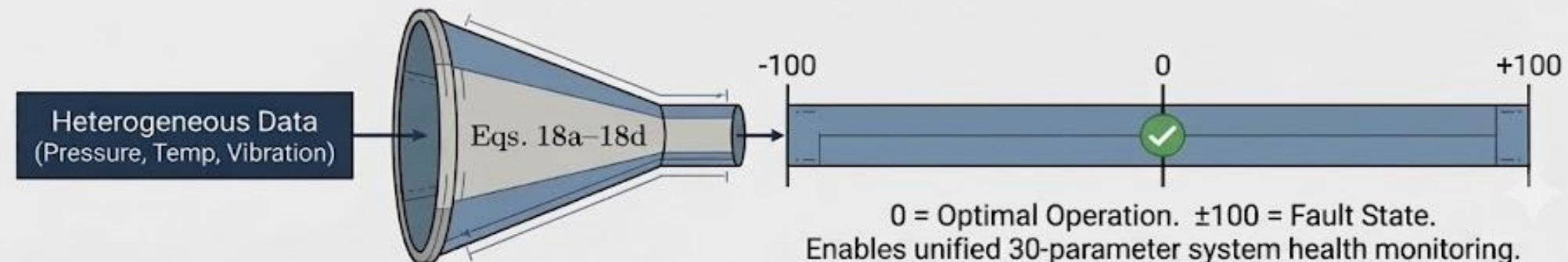
Linear Degradation Vector



$$y(t) = \alpha + \beta t + \epsilon(t)$$

Identifying critical degradation drift vs. stable noise.

The DIAG Normalisation Gauge





Experimental Study

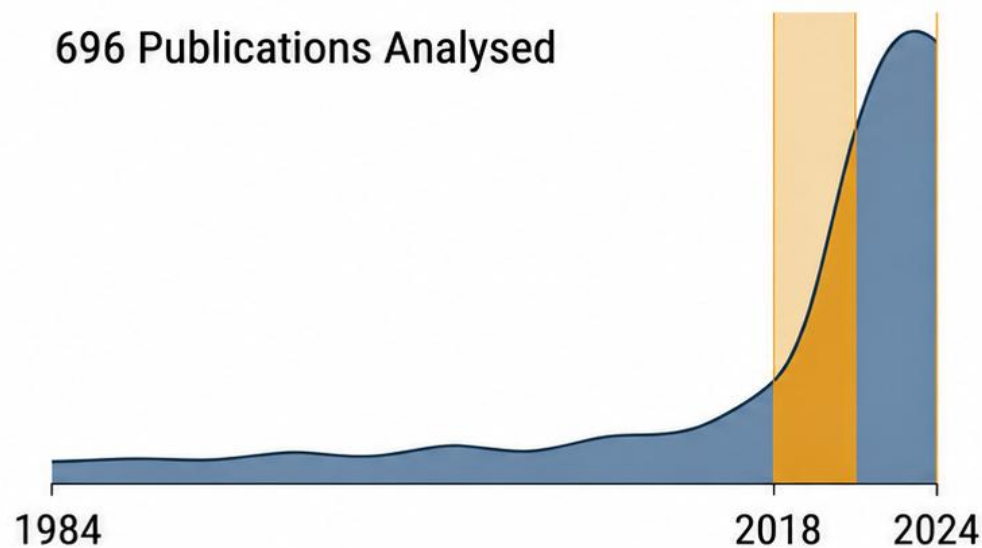


EXPERIMENTAL STUDY

Aviation: Deep Learning vs. Regulatory Transparency

The Macro View - SLR

696 Publications Analysed



Deep Learning Dominance: LSTM (98 papers) and CNN (108 papers) lead Remaining Useful Life (RUL) estimation post-2018.

The Micro View - Operations



The Regulatory Tension: High-accuracy Deep Learning models face a 4–5 year adoption lag due to strict EASA/FAA explainability requirements, driving a resurgence in logic-based expert systems.



Experimental Study



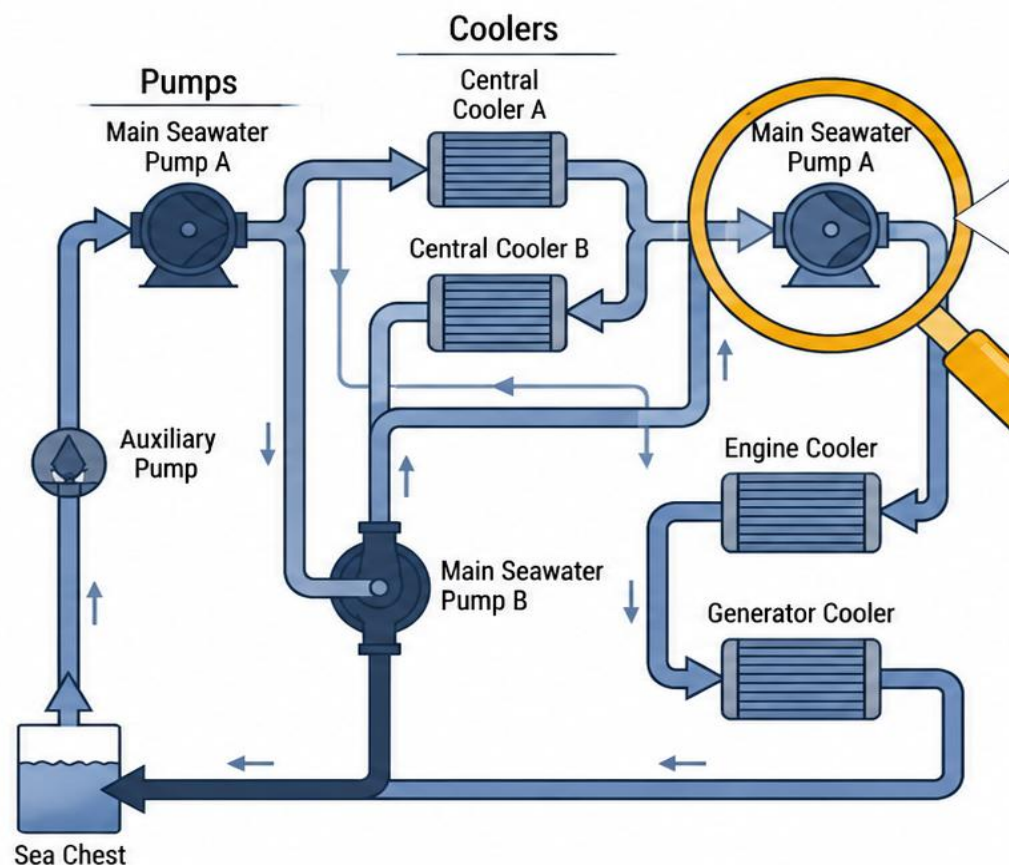
EXPERIMENTAL STUDY Maritime: Fault Arrays in Heavy Machinery

System Mapping

12 Functional States (F1-F12)

Spans single malfunctions (e.g., clogged chest) to complex compound faults.

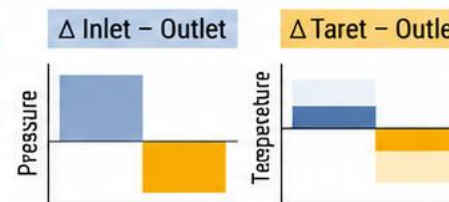
- F1: Normal Operation
- F2: Pump Cavitation
- F3: Clogged Sea Chest
- F4: Cooler Fouling
- F5: Valve Failure
- F6: Leakage
- F7: Pump + Cooler Compound
- F8: Sensor Drift
- F9: Flow Blockage
- F10: Temperature Spike
- F11: Pressure Drop
- F12: System Shutdown



The **DELTA** Innovation

$$\Delta = \text{Inlet} - \text{Outlet}$$

Algorithm calculates 10 DELTA parameters (pressure & temperature differentials).



Algorithm calculates 10 DELTA parameters (pressure & temperature differentials). This enriched signal differentiates overlapping fault modes far better than raw sensor telemetry, enabling direct integration into the Planned Maintenance System (LINK2PMS).



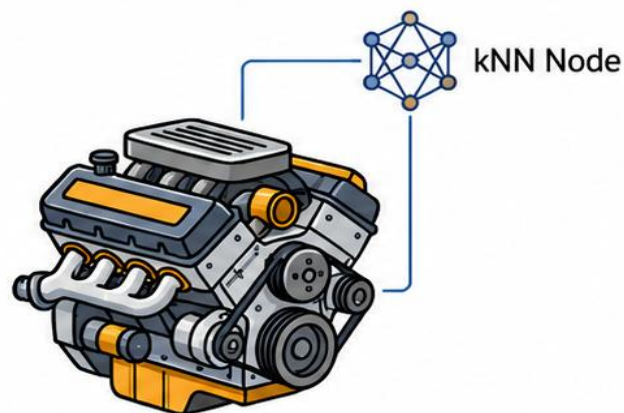
Experimental Study



EXPERIMENTAL STUDY

Automotive: Component Diagnostics & Manufacturing Forecasting

Component-Level: MOMENT System

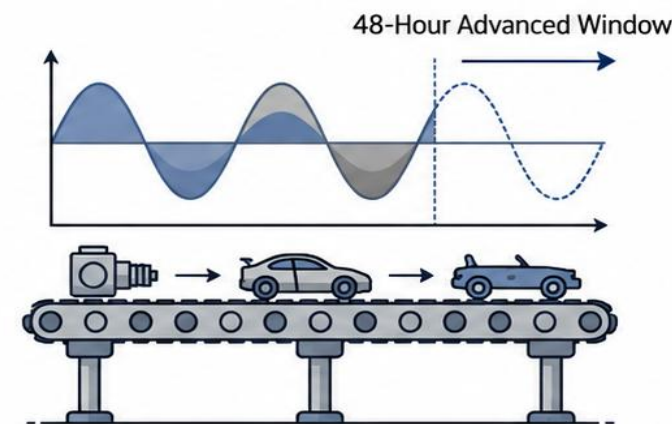


Dataset: 28,226 samples, 11 features.

Key Discriminators: Mass airflow & intake pressure.

Result: kNN achieves an F1-score of **0.983** for air filter health, outperforming SVM and RF.

System-Level: Manufacturing Plant



Dataset: 12 months of multi-sensor data (vibration, speed, temp).

Architecture: LSTM + Autoencoders for novel anomalies.

Result: LSTM achieves **0.12% Mean Absolute Error (MAE)** for 48-hour advanced failure forecasting.



Results



RESULTS

Cross-Domain Performance Matrix

Domain	Primary AI Models	Target System	Classification Outcome
Maritime	kNN	Cooling System	Real-time RUL monitoring
Aviation	LSTM, CNN, SageMaker	Flight Telemetry	Proactive Scheduling
Automotive (Component)	kNN	32,167 OBD-II samples	F1 = 0.983
Automotive (Plant)	LSTM, SVM, Autoencoder	Manufacturing Sensors	92% Accuracy

Universal Outcomes

>90%

Classification Accuracy across all sectors.

30%

Reduction in unplanned downtime.

25%

Global reduction in maintenance costs.



Results



RESULTS

Convergent Structural Challenges

Data Quality

Models are highly vulnerable to noisy, missing, or low-frequency sensor measurements. Garbage in, garbage out.

Interpretability

The 'Black Box' dilemma. High-performing Deep Learning architectures lack the transparent logic required for safety-critical diagnostics.

Regulatory Compliance

Strict aviation (FAA/EASA) and maritime auditability regulations demand manual explanations that purely statistical AI cannot natively provide.

Workforce Development

Mechanics and engineers require extensive, systematic retraining to trust, interpret, and action AI-generated probabilistic recommendations.



Concluding Remarks



CONCLUDING REMARKS

The Path Forward

Hybrid Architectures

Fusing the predictive power of Deep Learning with the auditable transparency of logic-based expert systems.

Federated Learning

Enabling secure, cross-fleet model training without centralising sensitive, proprietary operational data.

Multi-Platform Standardisation

Moving beyond single-vessel or single-chassis validations to create universal benchmark datasets.

AI-driven predictive maintenance is no longer just an operational advantage; it is an indispensable pillar of global transportation sustainability and safety.



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